

# Toward Sustainable Last-Mile Deliveries: A Comparative Study of Energy Consumption and Delivery Time for Drone-Only and Drone-Aided Public Transport Approaches in Urban Areas

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**Abstract**—The increasing demand for fast and energy-efficient parcel deliveries driven by the growth of e-commerce has made drones an attractive option for urban freight. However, considering drones' energy usage throughout the entire delivery process is essential when assessing their suitability. This research introduces an energy consumption model encompassing all flight stages—takeoff, flight, landing, hovering—simulating last-mile energy consumption within drone delivery-based systems. Additionally, it conducts a novel comparison between a system relying only on drones (Drone-only system) and a system combining drones with public transportation (Drone-APT system). The comparison considers factors such as consumer distribution, warehouse locations, and drone battery characteristics. The results, based on the real 3D structure of Barcelona, demonstrate that the Drone-APT system maintains consistent power consumption across different consumer distributions and warehouse locations. Conversely, the Drone-only system ensures consistent delivery times in these same scenarios. Furthermore, the Drone-APT improves energy efficiency, especially when direct drone delivery is limited by battery life. However, this efficiency improvement leads to increased delivery time due to the unpredictable timing of public transport vehicles. Striking a balance between drone energy consumption and delivery time is crucial, warranting further research to identify the most sustainable solution for last-mile delivery. This study offers valuable insights into the advantages and trade-offs associated with integrating drones and public transportation, laying the groundwork for future optimization efforts. Future work could incorporate intelligent technologies like machine learning to effectively address these trade-offs by dynamically optimizing drone paths based on energy efficiency and delivery time.

**Index Terms**—Drones, last-mile delivery, energy consumption, delivery time, public transport vehicles.

## I. INTRODUCTION

**L**AST-MILE parcel delivery operations encounter challenges in meeting the increasing demand for faster and more efficient deliveries. Companies like Amazon handle a massive volume of around 1.6 million packages per day (see [1]), resulting in significant costs for the last mile of delivery, as highlighted in a study conducted in [2]. The last mile stage can account for 20% to 50% of the total delivery cost. In the retail industry, the cost of the last mile represents approximately 30% of the product's value. Moreover, the use of commercial vehicles for deliveries contributes to greenhouse gas emissions and congestion on urban roads, both due to circulation and parking issues.

One effective approach to address these challenges involves leveraging drones. Reference [3], companies such as Amazon [4] and UPS [5] have been conducting tests and implementing drone-based delivery systems, which can be designed in various architectures. The first proposed architecture involves the use of Drones-only, lightweight drones that can fly directly from a warehouse or distribution center to the designated delivery location. However, these drones have limited battery life, which restricts their delivery range and payload capacity. Another approach is an architecture that combines the use of drones and trucks into a single delivery system (see [6]). In this approach, trucks act as a base station for drones, allowing them to deliver packages over short distances while trucks transport drones and packages over longer distances. While this approach offers numerous benefits, such as extending the overall range and reducing delivery time, it also involves additional costs associated with maintaining trucks and fuel expenses, along with undermining the initial focus of drone deployment on efforts to reduce CO<sub>2</sub> emissions if non-electrical vehicles are used.

To address these limitations, researchers have introduced an innovative framework designed to extend the delivery range of drones while eliminating the need for trucks [7], [8]. This approach capitalizes on the autonomous mobility of public

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transport vehicles. Drones are used in a way that allows them to land on the roofs of public vehicles, such as buses or trains, and travel alongside them to facilitate package delivery [9].

This system offers several advantages, particularly in its alignment with the growing demand for sustainable last-mile delivery solutions. Notably, it enhances energy efficiency and reduces downtown CO<sub>2</sub> emissions by leveraging existing transportation infrastructure. However, it introduces variability in delivery times due to the unpredictable schedules of public transportation vehicles, which can lead to inconsistency in delivery schedules [10]. Despite this drawback, the system creates potential job opportunities in related sectors, augmenting its overall benefits. Within our laboratory, we are actively collaborating with psychologists to delve into the acceptability of this system among individuals, aiming to comprehensively understand its adoption dynamics and user perceptions.

In most academic and scientific works discussing and improving Drone-based delivery systems, the primary technical factors considered for evaluation are typically energy consumption and delivery speed. However, these evaluations often rely on superficial energy consumption models that overlook certain stages of drone flight and fail to account for the three-dimensional urban environment. As a result, the energy consumption analysis is incomplete, leading to an unfair comparison of different Drone-based delivery systems.

In this paper, we address a significant gap in the evaluation of Drone-based delivery systems by developing an energy consumption model specifically tailored for last-mile deliveries in urban environments. Unlike existing models, our approach comprehensively considers all phases of drone flight, from takeoff to landing, and incorporates the complexities of dynamic 3D city environments. By doing so, we aim to provide a more accurate and fair comparison between different delivery approaches: Drone-only and Drone-APT. Overall, our contributions aim to advance the development of sustainable solutions for last-mile delivery and provide a comprehensive framework for evaluating the efficiency of Drone-based delivery systems in urban environments. Our contributions are as follows:

- **Comprehensive Energy Consumption Model:** We propose an energy consumption model that takes into account every aspect of drone flight during last-mile delivery operations, including takeoff, level flight, landing, and hovering. This model surpasses existing superficial models and provides a more nuanced understanding of energy consumption patterns.
- **Simulation in Dynamic Urban Environments:** Our model empowers us to simulate and analyze energy consumption within dynamic 3D city environments, incorporating factors such as customer distribution, warehouse locations, and drone battery characteristics. This comprehensive approach ensures that our evaluations accurately reflect real-world scenarios, providing valuable insights into the energy dynamics of last-mile delivery operations.
- **Investigation of Drone-APT System Sustainability:** We conduct a pioneering investigation into the sustainability of the Drone-APT system in terms of both power

consumption and delivery time. This analysis sheds light on the viability of this innovative approach and its potential impact on last-mile delivery efficiency.

- **Comparison of Delivery Systems:** We introduce a novel comparison between a last-mile delivery system based solely on drones and the Drone-APT system. By evaluating energy consumption and delivery time across these two approaches, we provide valuable insights into their respective strengths and weaknesses.
- **Contributions to Future Research:** Through our research, we offer valuable insights into the use of Drone-APT for last-mile delivery and open up new avenues for exploring the trade-offs between energy consumption and delivery time. Our work lays the groundwork for future studies in this emerging field.

The remainder of this paper is structured as follows. In Section II, we present a review of the relevant literature on drone-based last-mile delivery systems and provide a more detailed explanation of the motivation behind this study. Section III outlines our methodology for modeling the energy consumption of Drone-APT (Unmanned Aerial Vehicle-Assisted Public Transport) for delivery operations. Subsequently, in Section IV, we describe our simulation setup and assumptions. Section V is dedicated to the analysis and discussion of our results. Finally, in Section VI, we summarize our findings, acknowledge the limitations of our research, and propose suggestions for future work.

## II. LITERATURE REVIEW

In recent years, the use of unmanned aerial vehicles (UAVs), commonly known as drones, for last-mile deliveries has gained significant attention from both academia [8] and industry [4]. Although the utilization of drones for cargo transportation is not a new concept, recent technological advancements have significantly evolved it into a practical solution for addressing the increasing demand for rapid and efficient deliveries (refer to [11]).

Extensive research has been conducted to propose different architectures and models for efficient and sustainable Drone-based last-mile delivery systems. One of the early innovations was the introduction of the Drone-only delivery system, using autonomous drones for direct delivery from warehouse to customer [12], [13], [14], expediting the process with minimal human intervention. However, the inherent limitation lies in the battery life of these drones, constraining their range and rendering them unsuitable for covering longer distances efficiently.

To overcome this limitation, researchers have suggested novel frameworks integrating drones and trucks into a unified delivery system [15], [16]. In this configuration, the truck acts as a mobile depot, ferrying both drones and parcels to designated locations proximate to customers, from where the drones take-off and fly to final delivery point. This strategy offers numerous advantages, including expanded delivery coverage and decreased delivery duration. However, its principal drawbacks stem from elevated operational expenses attributed to truck driver salaries, upkeep expenditures, and fuel outlays.

Several studies have proposed a third architecture aimed at extending the delivery range beyond the initial design while avoiding the high costs associated with the second architecture, as demonstrated by research conducted by [8], [10], and [17]. The main idea is to use the mobility of public transport vehicles by transporting drones on their roofs to deliver packages. Although this approach offers cost and energy efficiency benefits, it introduces variability in delivery times due to the inherent uncertainties (such as delays and advances) associated with the use of a public transportation system.

Analyzing the potential advantages and disadvantages of drone-based delivery systems remains a vibrant area of research activity [18]. Recent studies have compared the total delivery costs of different systems, including conventional trucks, one-to-one drone delivery, drones supported by trucks, and one-to-many drone delivery [6]. The results show that drone deliveries can be a cost-effective alternative to traditional truck deliveries, but the specific circumstances and application should be considered. In addition, others researchers have compared the estimated costs per shipment of drones and SUGVs (Sidewalk Unmanned Ground Vehicles) for urban parcel delivery from various perspectives, such as practical, conceptual, technological, commercial, and environmental [19]. While SUGVs may be more profitable and acceptable from a commercial and resident's perspective, Drones have a better delivery range, especially when utilizing mother-ship delivery concepts, such as drones supported by trucks.

In the pursuit of developing sustainable delivery systems, many studies have looked on the energy consumption of drones during last-mile deliveries. For instance, [20] conducted a comprehensive comparison of the energy consumption and carbon emissions of drones with those of diesel and electric trucks. They found that a Drone-only parcel delivery system is generally not practical or energy-efficient. However, a hybrid system that combines drones with traditional delivery methods can be a more feasible and effective option. An innovative architecture was proposed, which designed a new vehicle-assisted drone delivery method that allows for multiple customers to be served in a single take-off while taking energy consumption into account. Comparative experiments proved the effectiveness of the proposed scheme [21].

Developing precise models to estimate drone energy usage represents a pivotal advancement in establishing sustainable delivery systems reliant on drones [27]. To this end, several studies have been conducted to explore different factors that affect drone energy consumption, such as delivery operation type, drone design, drone dynamics, and environment (see [28]). For example, [29] and [30] developed a model specifically designed for drones, utilizing the fundamental flight principles of manned aircraft. In contrast, [24] and [31] developed models based on helicopter operations that assume energy consumption during flight, takeoff, hover, or landing is approximately equivalent, while ignoring the effect of drone speed. To achieve greater accuracy on energy consumption estimation, [20] proposed a detailed three-component drone energy model that includes power to maintain lift, power to overcome parasite drag, and profile power to overcome

the rotational drag of the propeller blades. Furthermore, their model considers the idealized delivery process from a depot, with separate energy calculations for takeoff and ascent, steady level flight, descent, hovering, and landing for delivery.

Table I provides a comparison of research studies on drone-based delivery systems and factors influencing energy consumption and delivery time. The table compares these studies based on different system architectures (Drone-only, Drone-truck, Drone-APT), the drone flight phases considered in each study (straight, hovering, takeoff/landing), whether the paper provides an energy model or not, influential factors (including warehouse position, user distribution, and consideration of 3D environment), parameters of comparison used, and whether the work is tested in a real-world setting.

Based on this comparison and to the best of our knowledge, previous studies on drone-based delivery systems have utilized simplified or modified energy consumption models that fail to encompass all phases of drone flight, including landing and travel with vehicles, where drones consume minimal or no energy. However, the coordination and synchronization with paired vehicles may lead to significant hovering by the drones, resulting in increased power consumption. Additionally, these studies have typically assumed a two-dimensional urban environment, with fixed angles and altitudes for drone flight. This approach can lead to an incomplete analysis of energy consumption and an unfair comparison of different Drone-based delivery systems, especially when certain drone flight phases are more frequent in one system compared to another, as is the case with Drone-aided vehicles. Moreover, the required altitudes also differ between the two architectures, with drone-only systems having fixed altitudes while Drone-aided vehicles varying their altitudes based on the distance between two points on the delivery trip. Despite some recent efforts, there is still a significant lack of research that comprehensively compares the energy consumption and delivery time of drones between Drone-only and Drone-APT delivery systems.

Therefore, this paper proposes an energy consumption model that accounts for all phases of drone flight during last-mile delivery operations and simulates drone energy consumption in a three-dimensional urban area. Additionally, this work presents a pioneering comparison of energy consumption and delivery time between a Drone-only system and a Drone-APT system, considering factors such as customer distribution in the area, warehouse locations, and drone battery characteristics.

### III. ENERGY CONSUMPTION MODEL FOR DRONES IN LAST-MILE DELIVERY

To accurately estimate and analyze the power consumption in Drone-based delivery systems, this section introduces a new variant of an energy consumption model, inspired by the referenced work (see [20]). To enhance the comprehension of our model, we first outline a unified notation used throughout our modeling in SubSection III-A. We then delve into the configuration of steady flight, providing insights into the essential parameters and variables involved in Figure III-B. Expanding upon the steady configuration, we further explore Figure III-C, where we model the power demand during continuous flight

TABLE I  
COMPARISON OF RESEARCH STUDIES ON DRONE-BASED DELIVERY SYSTEM AND FACTORS INFLUENCING  
ENERGY CONSUMPTION AND DELIVERY TIME

| Ref.     | Systeme Architecture |             |           | Drone FlightPhase |          |                | Energy Model | Influential factors |                    |       | Parameters of comparison |                    | Real-world testing |
|----------|----------------------|-------------|-----------|-------------------|----------|----------------|--------------|---------------------|--------------------|-------|--------------------------|--------------------|--------------------|
|          | Drone-only           | Drone-truck | Drone APT | Straight          | Hovering | Takoff/landing |              | User distribution   | Warehouse position | 3D/2D | Delivery Time            | Energy consumption |                    |
| [22]     | X                    |             | X         | X                 |          |                | X            |                     |                    | 2D    | X                        | X                  |                    |
| [6]      | X                    | X           |           | X                 |          |                | X            |                     | X                  | 2D    | X                        | X                  |                    |
| [19]     | X                    |             |           | X                 |          |                | X            |                     |                    |       |                          | X                  |                    |
| [9]      |                      |             | X         | X                 | X        |                |              |                     |                    |       | X                        |                    |                    |
| [17]     |                      |             | X         | X                 | X        |                |              |                     |                    | 2D    | X                        |                    |                    |
| [23]     | X                    | X           |           |                   |          |                |              |                     |                    |       |                          |                    |                    |
| [24]     |                      | X           |           | X                 |          |                | X            | X                   |                    | 2D    | X                        | X                  | X                  |
| [25]     | X                    |             |           | X                 |          |                |              |                     |                    | 2D    | X                        |                    | X                  |
| [26]     |                      |             | X         | X                 | X        |                |              |                     |                    | 2D    | X                        |                    |                    |
| [20]     | X                    | X           |           | X                 |          | X              | X            | X                   |                    | 2D    |                          | X                  | X                  |
| Our work | X                    |             | X         | X                 | X        | X              | X            | X                   | X                  | 3D    | X                        | X                  | X                  |

across different flight phases in Drone-based last-mile delivery processes, considering the complexities of the urban 3D environment. To validate the accuracy and efficacy of our model, we present and discuss the results obtained from a comprehensive sensitivity analysis in Figure III-D.

#### A. Unified Notations

In this section, we provide a summary of the key parameters and coefficients used in the power model for drone delivery. These unified concepts serve as a foundation for constructing an accurate energy consumption model for drones. The parameters presented in this section were sourced from relevant literature [20], [32], and [33] and have been adjusted to align with the specific requirements of our study. By consolidating and organizing these parameters, we aim to facilitate a better understanding of the factors influencing energy consumption in drones during delivery operations (refer Table II).

#### B. Drone Steady Flight Configuration

For a detailed energy model, this paper adopts the approach used in previous studies (see [20], [33]), which calculates energy consumption based on the fundamental forces of flight. These forces include the weight force of the drone (due to gravity) and drag forces. Within this section, we will investigate the fundamental aerial forces that are involved in maintaining a stable flight of Drones. Figure 1 illustrates the coordinates and forces involved in the steady flight configuration of an drone.

As Figure 1 suggests, drones operating at a constant flight speed are subject to a variety of forces that influence their motion. These forces include the thrust force  $F_t$ , the air drag force  $F_d$  (comprising the combined effects of body drag  $F_{bd}$  and rotor drag  $F_{rd}$ ), the gravity force  $F_g$ , and the climbing force  $F_c$ .

$$F_t \sin \alpha = -F_d - F_g \cdot \sin \beta \quad (1)$$

$$F_t \cos \alpha = F_g \cos \beta \quad (2)$$

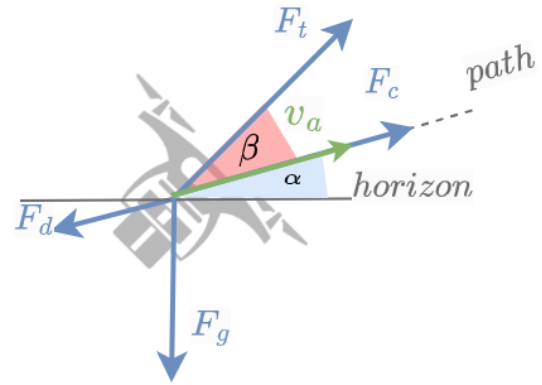


Fig. 1. Coordinate and forces in steady flight configuration of a drone.

In a state of trimmed flight, where a drone maintains a specific airspeed and flight path angle, the required thrust force can be calculated using equations 1 and 2 as follows:

$$F_t = \sqrt{m^2 g^2 + F_d^2 + 2 F_d \cdot F_c} \quad (3)$$

where  $F_c$ ,  $F_g$ ,  $F_{bd}$ , and  $F_{rd}$ , are calculated as follows:

$$F_c = F_g \cdot \sin \alpha \quad (4)$$

$$F_g = g \cdot \sum_{k=0}^3 m_k \quad (5)$$

$$F_{bd} = \frac{1}{2} \cdot \rho \cdot v^2 A \cdot c_{air} \quad (6)$$

$$F_{rd} = \rho \cdot R \cdot v_t^3 \cdot \left( 1 + 2 \cdot \frac{v}{v_t^2} \right) \cdot \frac{\sigma \cdot c_{bd}}{n_{rotors}} \quad (7)$$

where the speed of blade  $v_t$  is calculated as follows [33]:

$$v_t = \sqrt{\frac{6 \cdot m \cdot g}{n_{rotors} \cdot n_{blades} \cdot \bar{c} \cdot \bar{c}_l \cdot \rho \cdot r}} \quad (8)$$

TABLE II  
NOTATION FOR DRONE ENERGY DEMAND MODELING

|             |                                    |                            |                                                 |
|-------------|------------------------------------|----------------------------|-------------------------------------------------|
| $F_t$       | thrust force                       | $\rho$                     | air density( $1.225\text{kg/m}^3$ )             |
| $F_{bd}$    | body drag force                    | $g$                        | gravity constant( $9.81\text{kg/s}^2$ )         |
| $F_{rd}$    | rotor drag force                   | $v$                        | Drone speed( $0 - 22\text{m/s}$ )               |
| $F_g$       | gravity force                      | $v_t$                      | blade tips speed                                |
| $F_c$       | climbing force                     | $A$                        | front surface area of Drone( $0.15\text{m}^2$ ) |
| $P^{climb}$ | power required for climbing        | $r$                        | rotor radius( $0.4\text{m}$ )                   |
| $P^{lift}$  | power required for lift            | $R$                        | rotor disc area( $0.8\text{m}^2$ )              |
| $P^{BDrag}$ | power to overcome body drag        | $n_{rotors}$               | numbers rotors( $8\text{rotors}$ )              |
| $P^{RDrag}$ | power to overcome rotor drag       | $n_{blades}$               | numbers blades( $3\text{blades}$ )              |
| $P^{int}$   | power losses of internal factors   | $c_{bd}$                   | blade drag( $0.075$ )                           |
| $m_{Drone}$ | Drone weight( $3.8\text{kg}$ )     | $c_{air}$                  | air drag( $0.3$ )                               |
| $m_{pay}$   | payload weight( $0 - 4\text{kg}$ ) | $\sigma$                   | rotor solidity ration( $2.51/\text{m}$ )        |
| $m_{bat}$   | battery weight( $6 - 8\text{kg}$ ) | $\bar{c}$                  | rotor mean chord ( $0.1$ )                      |
| $\alpha$    | flight path angle                  | $\bar{c}_l$                | blade lift ( $0.4$ )                            |
| $\beta$     | rotor angle of attack              | $\epsilon^{Drone}$         | transmission efficiency( $0.93$ )               |
| $k$         | lifting power markup ( $1.15$ )    | $\epsilon_{charg}^{Drone}$ | charging efficiency( $0.93$ )                   |
| $w$         | down wash coefficient              |                            |                                                 |

Now, that we have defined the different forces that influence drone motion, we can express the total power required by the drone to overcome these forces:

$$P^{Drone} = P^{lift} + P^{climb} + P^{BDrag} + P^{RDrag} + P^{int} \\ = k \cdot w \cdot F_t + v \cdot F_c + v \cdot F_{bd} + v \cdot F_{rd} + P^{int} \quad (9)$$

where  $P^{lift}$  is the power needed to generate lift,  $P^{int}$  represents the power losses due to internal factors (e.g., motor efficiency),  $P^{climb}$  is the power required for climbing or descending,  $P^{BDrag}$  and  $P^{RDrag}$  represents the power to overcome body drag and rotor drag respectively.

### C. Modeling of Power Demand During Repeated Flight Phases

To achieve an accurate model of the energy consumption in a Drone-APT system, it is crucial to consider the impacts of the heights and shapes of the overflown buildings on the repeated flight phases of drones, such as takeoff, landing, and correspondences between public transportation vehicles. Indeed, an urban landscape requires many adjustments to the flight angle, speed, and altitude to navigate safely and efficiently. By accounting for these factors, a more comprehensive understanding of energy consumption can be obtained.

Figure 2 illustrates the last-mile delivery process of drone using public transport vehicles (buses) over an urban building, showcasing different flight stages and 3D measurements scaling. To estimate the power demand accurately, we use 9 and weight it based on the associated duration, total weight of the drone, different flight angles, and drone speed. Through the analysis of specific flight angles over time, we can accurately determine the energy demands of the drone delivery process as a function of time.

$$E^{Drone}(t, v, m, \alpha) = \frac{1}{\epsilon^{Drone} \cdot \epsilon_{charg}^{Drone}} \cdot t \cdot P^{Drone}(m, v, \alpha) \\ + t \cdot \frac{P^{int}}{\epsilon_{charg}^{Drone}} \quad (10)$$

The cumulative energy required for the drone delivery process during the duration from 0 to T (the desired delivery

time) can be computed by integrating the energy demand function (refer to 10) over the specified time interval in the following manner:

$$E_{total}^{Drone}(T, v, m, \alpha) = \int_0^T E^{Drone}(t, v, m, \alpha) dt \quad (11)$$

### D. Sensitivity Analysis

Here, we present a sensitivity analysis conducted to examine the energy consumption of drones. The objective of this analysis is to evaluate how different factors influence energy efficiency and overall power demand. By systematically varying parameters such as vehicle speed, payload weight, and environmental factors (such as building height and positions), we can gain valuable insights into the sensitivity of energy consumption to these variables.

To ensure the accuracy and practical applicability of our energy consumption model to real-world scenarios, we adopted the technical parameters of the Amazon Prime Quadcopter Air delivery drone as a reference [32]. These specifications serve as a reliable basis for setting the key characteristics of the Drone in our analysis. The detailed technical parameters of the Amazon Prime Quadcopter Air Delivery drone are provided in Table III.

By applying our mathematical energy model to a drone in a steady flight configuration, as illustrated in Figure 1, we compute the variations in the total energy consumption of the drone resulting from changes in the angle of flight, drone speed, and total payload. The outcomes of this analysis are presented in Figure 3.

The figure shows three distinct analyses: the impact of the angle of flight on energy demand (Figure 3a), the influence of drone speed on energy demand (Figure 3b), and the relationship between total payload capacity and energy demand (Figure 3c).

In Figure 3a, the variation of the drone's total energy demand is observed by manipulating the angle of flight. It is remarkable to observe that at an angle of  $-90^\circ$ , representing the vertical landing stage, the energy consumption is equivalent to 1/4 of the energy consumed during the vertical take-off

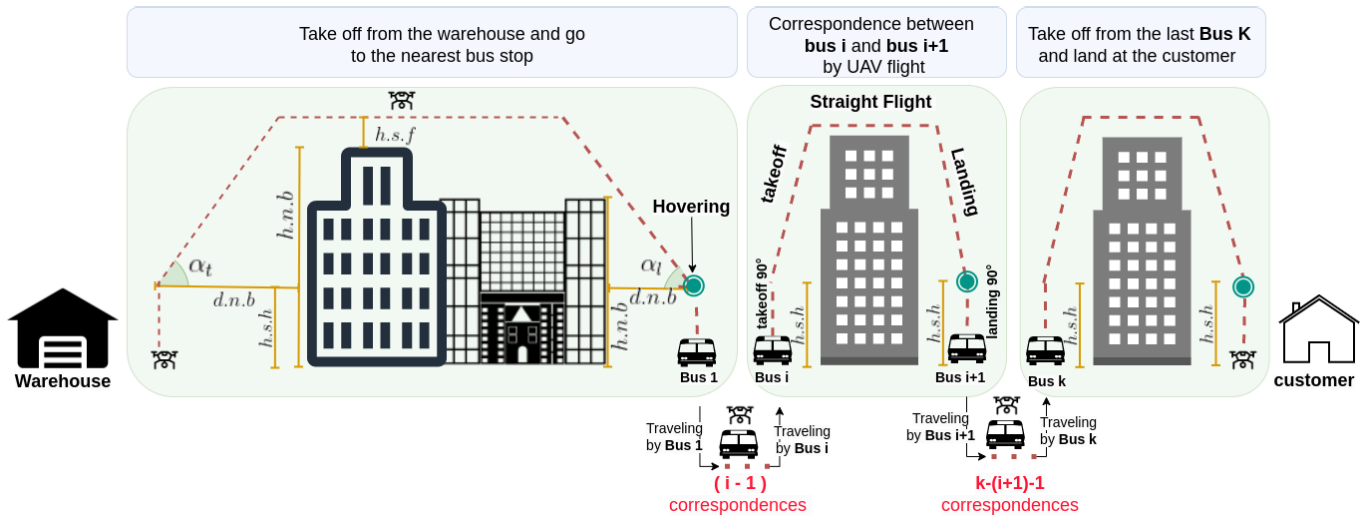


Fig. 2. Last-mile delivery path of drone using public transport vehicles (buses) over an urban building, with different flight stages and 3D measurements scaling.

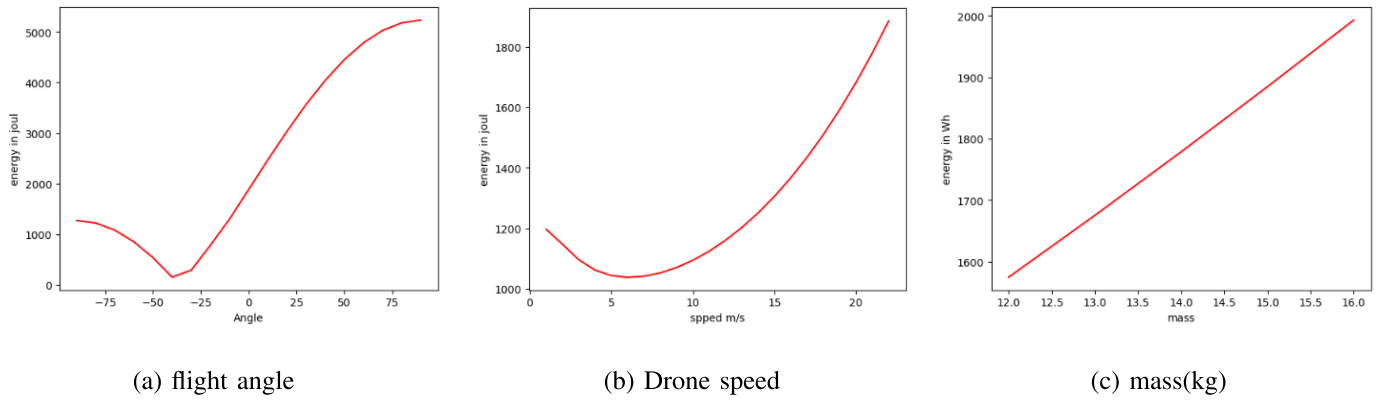


Fig. 3. Energy consumption of drone for steady flight configuration by different flights angles, drone speeds, and mass.

TABLE III  
AMAZON PRIME QUADCOPTER AIR DELIVERY TECHNICAL PARAMETERS

| Specification Parameter | Value                                                            |
|-------------------------|------------------------------------------------------------------|
| Drone Model             | Amazon Prime Quadcopter Air Delivery                             |
| Drone Weight            | 3.8 kilograms                                                    |
| Max. Package Weight     | 2.3 kilograms                                                    |
| Max. Payload Capacity   | 16 kilograms                                                     |
| Max. Flight Time        | 30 minutes                                                       |
| Max. Speed              | 97 km/h                                                          |
| Max. Altitude           | 122 meters                                                       |
| Battery Type            | Lithium polymer (LiPo 6S/10S 10,000 mAh)                         |
| Charging Time           | Approximately 2 hours                                            |
| Operating Range         | 16 km                                                            |
| Dimensions              | Length: 0.91 meters<br>Width: 0.91 meters<br>Height: 0.30 meters |

stage at  $90^\circ$ . Conversely, during direct flight (at 0 degrees), the energy demand falls between the values observed during the landing and take-off stages.

By analyzing the Figure 3(b), we gain a deeper understanding of how different speeds impact the drone's energy consumption. Notably, as the speed of movement increases from 5 m/s to 20 m/s, the energy consumption also increases accordingly. Conversely, as the speed increases from 0 to 5 m/s, there is a decrease in energy consumption. This is a result of transitioning from the hover stage to straight flight, where low-speed straight flight consumes less energy than hovering to maintain a stable position.

Finally, upon analyzing Figure 3(c), we can examine the correlation between payload weight and energy consumption. It is evident that as the payload weight increases, the total energy consumed logically increases as well.

These results highlight the significance of taking into account flight angles, speed, and payload weight in accurately modeling energy consumption in drone operations. Incorporating these factors allows for the development of more realistic energy models, enabling a fairer comparison among various last-mile delivery systems.

#### IV. GENERAL ASSUMPTIONS AND EXPERIMENTAL METHODOLOGY

Within this section, we introduce the framework and methodology used for comparing the energy consumption

TABLE IV  
SUMMARY OF ASSUMPTIONS FOR SIMULATION

|                                                                |                                                 |
|----------------------------------------------------------------|-------------------------------------------------|
| Drones have limited battery life                               | Recharging of drones is not considered          |
| Drone can only serve one parcel by trip                        | Cargo have the same type                        |
| Cargo has the same weight for both systems                     | Drones can fly directly                         |
| Drones can land/takeoff on the roof of a bus                   | Drones can wait for the bus at the bus station  |
| Drones can use several buses on their trips to reach customers | Traffic conditions can change during a bus trip |
| Wind conditions are neglected for drones trips                 | drones can fly at different altitudes           |

and delivery time of drones across various delivery systems. In SubSection IV-A, we outline the general assumptions made during the analysis to ensure transparency and clarity. Additionally, in SubSection IV-B, we provide an overview of the experimental methodology employed to effectively quantify and compare the energy consumption and delivery time of different Drone-based delivery systems.

#### A. General Assumptions

This section outlines a simulation study that compares parcel delivery services utilizing Drone-only and Drone-aided public transportation (Drone-APT). To make the comparison, a set of assumptions is made, which are summarized in Table IV. In subsequent computational experiments, we considered the real 3D structure of the city of Barcelona. We used the OpenStreetMap framework (see [34]) to obtain the location of the different buildings, their heights, and the positions of bus stations, as well as the road network with different road types. We also utilized the Blender package (see [35]) for 3D geospatial visualization, analysis, and exploration. Figure 4 shows the 3D geospatial visualization of Barcelona. Furthermore, we used [36] to obtain real-time transit data of Barcelona city for planning the delivery path of drones through public transit vehicles with real-time data. The delivery time was chosen to be between 11 am and 2 pm to capture the peak period of high crowding and its impact on the movement of public transport vehicles. This timeframe was selected as a worst-case scenario for Drone-APT measurements due to the challenging conditions posed by heavy traffic and congestion during that time.

#### B. Experimental Methodology

To ensure a consistent and reliable comparison between Drone-only and Drone-APT tours during the delivery process, specifically concerning energy consumption and delivery time, it is crucial to account for various factors that can impact the outcomes. These factors encompass the distribution of consumers within the area being served, the strategic placement of warehouses, and the specific characteristics of the drone batteries being used. By taking these factors into account, a more rigorous and scientific comparison can be achieved. To this end, we implemented different scenarios with different consumers' distributions over the area being served, varied warehouse positions, and use different types of drone batteries.

1) *Consumers Distribution in the Area:* In our study, we employed various user distribution models, including uniform, normal, and cluster distributions, to account for the

different population distribution typologies found in various cities. These models allowed us to gain valuable insights into how the Drone-based delivery systems performed under diverse user distribution patterns, leading to a more precise evaluation of their capabilities and effectiveness. As shown in Figure 5, we presented three distribution models that depicted the spatial distribution of 10 consumers in Barcelona, serving as illustrative examples of the scenarios considered in our analysis.

a) *Uniform distribution:* In practicality, a uniform distribution of users in the context of delivery represents a situation where the demand for deliveries is evenly distributed throughout various areas of the city. In our simulation of Barcelona city with 10 consumers, we have incorporated a uniform distribution model to evenly disperse the consumers across the entire city. Whereby, the city was divided into equal-sized grid cells, and an equal number of consumers were assigned to each cell. To introduce some randomness within the uniform distribution, the coordinates for each consumer were generated randomly within the boundaries of their respective grid cell (see Figure 5a). Whereby, the position ( $x_{\text{coordinate}}$ ,  $y_{\text{coordinate}}$ ) for each consumer can be calculated as:

$$(x_{\text{coordinate}}, y_{\text{coordinate}}) = (r \cdot W + i \cdot W, r \cdot H + j \cdot H) \quad (12)$$

Here,  $W$  (resp.  $H$ ) represents the width (resp. height) of the grid cell,  $r$  is a random number between 0 and 1,  $i$  (resp.  $j$ ) is the grid cell index in the x-direction (resp. y-direction).

b) *Normal distribution:* In the real-world context of delivery systems, a normal distribution of users accurately represents the natural pattern of demand, with concentrations of users in specific areas of the city. To simulate this scenario in our Barcelona city simulation with 10 consumers, we employed a normal distribution model to distribute the consumers across the city. Each consumer's position, denoted as ( $x_{\text{coordinate}}$ ,  $y_{\text{coordinate}}$ ), was generated using a normal distribution with mean  $\mu$  and standard deviation  $\sigma$  (see Figure 5b). The calculation is as follows:

$$(x_{\text{coordinate}}, y_{\text{coordinate}}) = (\mu_x + Z_x \cdot \sigma_x, \mu_y + Z_y \cdot \sigma_y) \quad (13)$$

Here,  $Z_x$  and  $Z_y$  are random values generated from a standard normal distribution, while  $\mu_x$  and  $\mu_y$  represent the mean values of the  $x$ -coordinate and  $y$ -coordinate, respectively. Additionally,  $\sigma_x$  and  $\sigma_y$  indicate the standard deviations of the  $x$ -coordinate and  $y$ -coordinate, respectively.

c) *Cluster distribution:* In practical delivery scenarios, the distribution of users often exhibits clustering behavior, where the demand for deliveries is concentrated in several distinct areas of the city. To simulate this clustering effect in our

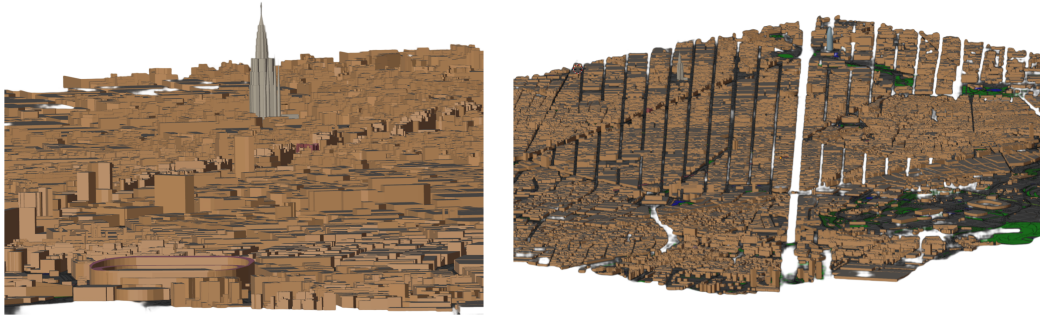


Fig. 4. The 3D geospatial visualization of Barcelona city.

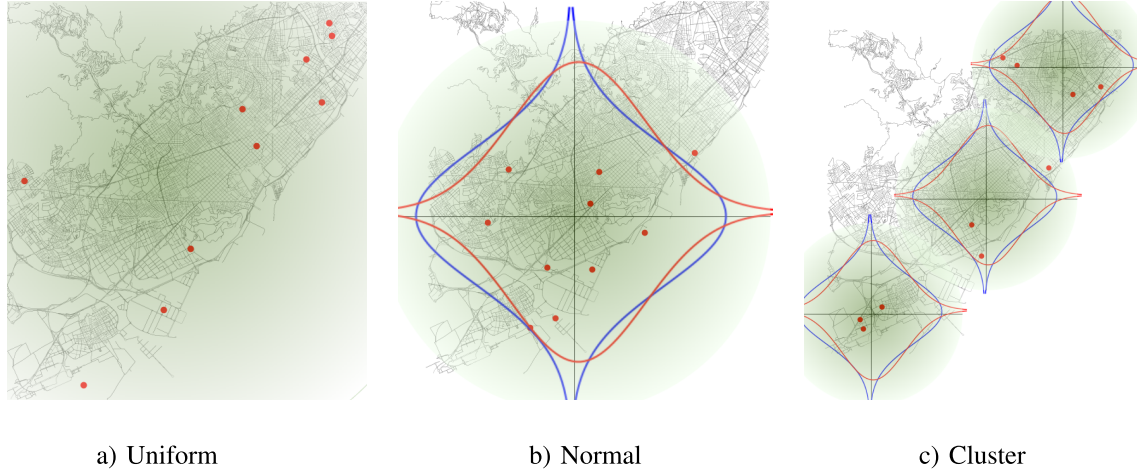


Fig. 5. Different distribution models for 10 consumers in Barcelona.

Barcelona city simulation with 10 consumers, we employed a model using several Normal distributions to distribute the consumers across the city. Each consumer's position, denoted as  $(x_{\text{coordinate}}, y_{\text{coordinate}})$ , was generated by combining multiple Normal distributions, each representing a distinct cluster (see Figure 5c). The coordinates can be calculated using the following equation:

$$(x_{\text{coordinate}}, y_{\text{coordinate}}) = (\mu_{x_i} + Z_{x_i} \cdot \sigma_{x_i}, \mu_{y_i} + Z_{y_i} \cdot \sigma_{y_i}) \quad (14)$$

Here,  $Z_{x_i}$  and  $Z_{y_i}$  are randomly generated values from a standard normal distribution for each cluster  $i$ . The cluster centers  $(\mu_{x_i}, \mu_{y_i})$  represent the mean values of the  $x$ -coordinate and  $y$ -coordinate for cluster  $i$ , respectively, while  $(\sigma_{x_i}, \sigma_{y_i})$  indicate the standard deviations of the  $x$ -coordinate and  $y$ -coordinate for cluster  $i$ , respectively.

2) *Warehouse Positions*: The choice of warehouse position in drone delivery systems is a critical factor that requires careful consideration, taking into account various factors such as proximity to pickup and delivery locations, energy consumption, delivery time, and cost. Numerous studies have explored the impact of warehouse position on system performance and have provided valuable insights (e.g., see [37]). In our comparative analysis of Drone-only and Drone-APT systems, we aim to align with different strategic positioning strategies for warehouses. To achieve this, we evaluate the performance of both systems across three distinct warehouse

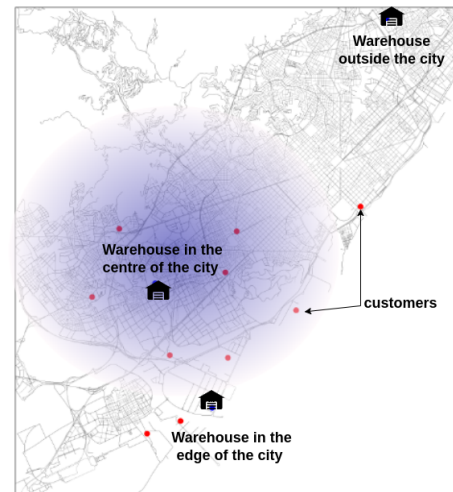


Fig. 6. The three different warehouse positions in Barcelona.

positions: in the city center, on the edge of the city, and outside of the city as depicted in Figure 6.

3) *Drone Delivery Batteries Features*: The choice of battery type plays a crucial role in determining the performance and capabilities of delivery systems. For that, several studies (e.g., [38], [39]) have investigated the characteristics and features of various battery types for Drone delivery applications. These references provide valuable insights into the energy capacity,

TABLE V  
COMPARISON OF DIFFERENT DRONE BATTERY TYPES

| Battery Type                 | Capacity (mAh) | Voltage (V) | Flight Time (min) | Ref. or Battery Manufacturers |
|------------------------------|----------------|-------------|-------------------|-------------------------------|
| Nickel Metal Hydride (NiMH)  | 8500           | 7.4         | 40-45             | [40]                          |
| Lithium Ion (Li-Ion)         | 8400           | 7.4         | 45-50             | [40]                          |
| Lithium-Ion Polymer (LiPo)   | 10000          | 22.4        | 30-35             | [32]                          |
| Lithium Polymer battery pack | 20000          | 6-22.2      | 45-90             | [41]                          |
| Li-Po Battery Pack           | 40000          | 22.8V       | long time         | [42]                          |

flight time, and other relevant parameters of these battery types. In our comparative study, we consider the use of different battery types to assess their impact on system performance. Table V provides a comparison of various battery types used in drones for delivery. The table includes information on the battery type, capacity (in milliampere-hours - mAh), voltage (in volts - V), estimated flight time (in minutes), with actual references to the respective battery manufacturers or research papers.

## V. SIMULATION RESULTS AND ANALYSIS

In this section, we conduct comprehensive simulations to evaluate and compare the performance of Drone-based delivery systems, specifically Drone-only, Drone-APT, and Hybrid (Drone-only/APT).

We reference the technical parameters of the drone used in our simulation to those of the Amazon Prime Quadcopter Air delivery drone [32]. These specifications provide a dependable foundation for establishing the key characteristics of the drone in our analysis. Detailed technical parameters of the Amazon Prime Quadcopter Air Delivery drone can be found in Table III. Firstly, in the Drone-only system, the drone is intended to make direct flights between the warehouse and customers at different altitudes depending on the buildings' heights within a 3D model of Barcelona city. The drone travels at a fixed speed of 20 m/s. Secondly, for the Drone-APT system, we utilize TMB GTFS data [43] to determine the path from the warehouse to the customer by bus. We then establish direct drone flight links between the warehouse and the first bus stop, the last bus stop, and the customer, as well as between bus stops if a correspondence exists, adjusting altitude based on the buildings encountered along the way. Finally, the hybrid system combines the strengths of both the Drone-only approach and Drone-APT to enhance last-mile delivery efficiency. By integrating direct flight segments using drones and travel segments via public transportation vehicles, with a percentage split of 40-60% within the same delivery path, we aim to optimize delivery operations.

We explore a total of 24 distinct scenarios outlined in SubSection IV, encompassing different battery types, including Li-Ion, LiPo Pack v1, and LiPo Pack v2 (see Table V). Additionally, we consider three warehouse positions: center, edge, and outside, as well as three distribution models:

TABLE VI  
SIMULATION PARAMETERS

| Parameter                                | Value              |
|------------------------------------------|--------------------|
| Drone speed                              | 20m/s              |
| Payload capacity                         | 10kilograms        |
| Simulation area                          | 100km <sup>2</sup> |
| Number of customers                      | 10                 |
| Number of warehouses                     | [1, 3]             |
| Customer distribution variance (normal)  | 5km                |
| Customer distribution variance (cluster) | 2km                |
| Safe fly altitude                        | 10m                |

uniform, normal, and clusters. Through rigorous analysis, we delve into the outcomes in-depth. To assess and contrast the performance of each system, we use two important metrics: energy consumption and delivery time. Table VI provides the simulation parameters used in our comparison.

### A. Energy Consumption Metric

Figure 7 clearly demonstrates that Drone-only exhibits significantly higher energy consumption compared to Drone-APT in all scenarios, while the energy consumption of the hybrid system (Drone-only/APT) falls between the two.

Moreover, it is noteworthy that Drone-APT remains unaffected by the distribution of consumers or the location of the warehouse. It maintains a consistent power consumption of approximately 7000 mAh across all scenarios, and there are no discernible differences in energy consumption among the 10 consumers within each scenario.

On the other hand, Drone-only is influenced by the position of the warehouse. Specifically, when the warehouse is centrally located within the city, it consumes significantly more energy to scenarios where the warehouse is situated on the city's edge (in the context of normal distribution). Similarly, the distribution of consumers also impacts the energy consumption of Drone-only. For instance, in scenarios with cluster distribution, the energy consumption ranges from 15000-20000 mAh, while in scenarios with normal distribution, it increases to 25000-30000 mAh when the warehouse is located in the center of the city. Furthermore, Drone-only exhibits variations in power consumption among the 10 consumers, particularly when the warehouse is located outside the city.

Lastly, the hybrid system (Drone-only/APT) generally falls between the two systems in most scenarios. However, there is a noticeable difference when the distribution of consumers follows the normal distribution and the warehouse is located outside the city. This can be attributed to the nature of the algorithm used to balance delivery time and energy consumption, as well as the stochastic nature of the public transport network's timing.

### B. Delivery Time Metric

As shown in Figure 8, it is evident that Drone-only consistently achieves significantly lower delivery times compared to

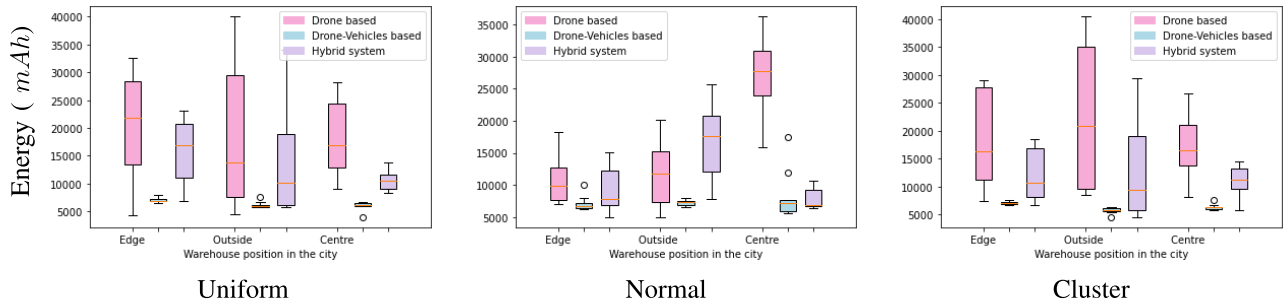


Fig. 7. Energy consumption comparison (by mAh) by warehouse position and customer distribution in the area.

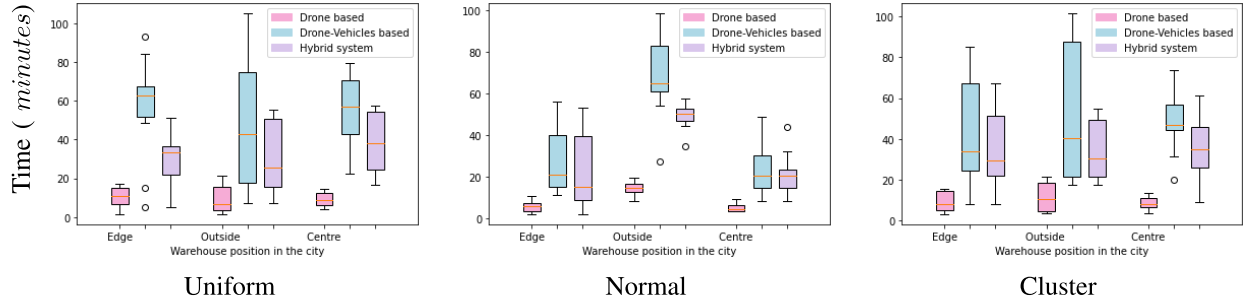


Fig. 8. Time comparison (by minutes) by warehouse position and customer distribution in the area.

Drone-APT across all scenarios. The hybrid system (Drone-only/APT) exhibits delivery times that fall between the two systems.

Additionally, it is worth mentioning that the delivery time of Drone-only is not impacted by the distribution of consumers or the location of the warehouse. It consistently achieves a delivery time of less than 20 minutes in all scenarios, and there are no noticeable variations in delivery time among the 10 consumers within each scenario.

On the contrary, the delivery time of Drone-APT is influenced by the warehouse position. Specifically, when the warehouse is located centrally within the city, the delivery time to the consumers is shorter, ranging from 15 to 30 minutes (in the case of normal distribution). In contrast, scenarios where the warehouse is situated on the outside of the city result in longer delivery times, ranging from 60 to 80 minutes (in the case of normal distribution). Furthermore, the distribution model of consumers also affects the delivery time of Drone-APT. For instance, in scenarios with cluster distribution, the delivery time ranges from 45 to 55 minutes, whereas in scenarios with normal distribution, it decreases to 15 to 30 minutes when the warehouse is positioned in the city center. Moreover, Drone-APT displays variations in delivery time among the 10 consumers, particularly when the warehouse is located outside the city.

Lastly, the hybrid system (Drone-only/APT) generally exhibits intermediate delivery times compared to Drone-only and Drone-APT in all scenarios.

### C. Battery Percentage Comparison in Different Scenarios

Figure 9 presents a comparative analysis of the battery percentage based on different battery types, customer distribution models in the area, and warehouse positions. The objective is to evaluate the influence of these factors on the battery

consumption during drone delivery operations. Additionally, this analysis helps determine whether the energy consumed exceeds the total battery capacity, thus aiding in selecting the appropriate battery type for each system and scenario.

The outcomes reveals that the percentage power consumption of the Li-Ion batteries surpassed 100 percent in all scenarios within the Drone-only and hybrid system, and ranged between 80-100 percent in Drone-APT. This indicates that the use of Li-Ion batteries resulted in incomplete delivery operations in most cases. In contrast, the Li-Po battery pack version 2 exhibited a battery percentage that did not exceed 50 percent in all scenarios. This implies that the delivery process can be successfully completed using this type of battery.

In contrast, when the warehouse is located in the center of the city and a LiPo battery Pack version 1 is used, the percentage of energy consumption differs across the systems. This difference is also observed when the warehouse is located in the edge or outside of the city, and for both uniform and cluster distributions. Specifically, both the hybrid system and Drone-APT exhibit a percentage of energy consumption around 50%. However, in the case of Drone-only, the percentage approaches and exceeds 100%. While for normal distribution the percentage is around 50%.

### D. Discussion and Analysis of Systems Performance

The simulation results provide invaluable insights into the performance metrics of diverse delivery systems, covering aspects such as energy consumption, delivery times, and battery utilization. The most important findings from the study are outlined as follows:

#### • Energy Consumption Analysis:

- Drone-only exhibits notably higher energy consumption compared to Drone-APT across all scenarios,

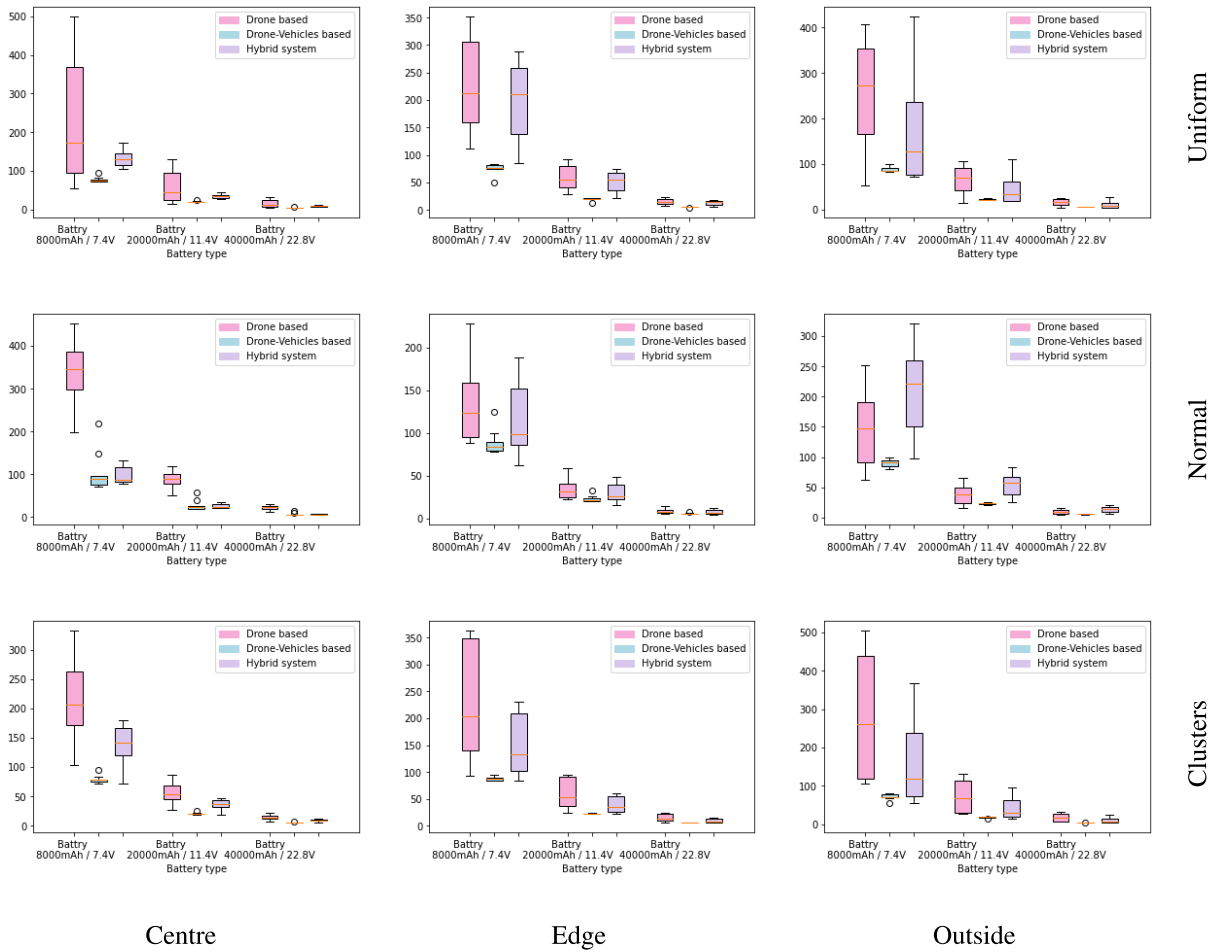


Fig. 9. Percentage battery consumption comparison by type of battery, customer distribution in the area, and warehouse position.

indicating the influence of system architecture on energy efficiency.

- The geographic placement of warehouses and the distribution of consumers significantly impact power usage in the Drone-only model, underscoring the necessity of considering these factors when optimizing drone energy consumption.

#### • Delivery Time Evaluation:

- Drone-only consistently achieves faster delivery times compared to Drone-APT in all scenarios.
- Warehouse location and consumer distribution exert a significant influence on delivery time within the Drone-APT system, necessitating their consideration during optimization efforts.

#### • Drone-APT System Dynamics:

- Drone-APT maintains uniform power consumption across diverse consumer distributions and warehouse placements due to its integration with public transport, which mitigates the need for extensive flight distances.

#### • Battery Selection Implications:

- The selection of battery type significantly influences both battery percentage and operational capacity. Opting for higher-capacity options such as LiPO Pack v2 enables the successful completion of delivery tasks,

albeit at the expense of increased costs and payload weight.

#### • System Hybridization Insights:

- The hybrid system (Drone-only/APT) represents a middle ground between the standalone drone models, offering a compromise solution that balances energy efficiency and delivery time. The hybrid system's effectiveness may vary based on specific operational contexts.

In summary, the most significant findings encompass the substantial differences in energy consumption and delivery times between the Drone-only and Drone-APT systems, the critical role of battery selection in operational feasibility, and the potential benefits of hybridizing delivery systems to achieve a balance between efficiency and speed. These insights underscore the importance of considering various factors when designing and optimizing drone delivery systems.

## VI. CONCLUSION AND FUTURE WORK

In conclusion, this study investigates the energy consumption and delivery performance of systems based on drones in the context of last-mile parcel delivery. The proposed energy consumption model considers all phases of drone flight and compares the performance of a system that combines drones

and public transportation vehicles (Drone-APT) to a system that solely relies on drones (Drone-only). The analysis reveals that Drone-APT exhibits higher energy efficiency compared to Drone-only across all scenarios. Furthermore, Drone-APT maintains consistent power consumption regardless of consumer distributions and warehouse locations. However, it is important to note that Drone-APT experiences longer delivery times due to potential delays associated with public transportation. The choice of battery type is crucial as it impacts battery capacity, successful completion of delivery operations, and overall system costs. The findings highlight the potential of a hybrid system, combining the advantages of Drone-only and Drone-APT, to strike a balance between energy efficiency and delivery time. This hybrid system warrants further optimization through the exploration of different strategies and algorithms.

In considering the limitations of the current analysis, it's essential to acknowledge that our study focused on a specific set of scenarios and assumptions, which may not fully capture the complexities of real-world conditions. Variations in weather, airspace regulations, and technological advancements could impact the outcomes observed in this study. Therefore, as next step, research should aim to address these limitations by conducting more extensive field experiments and incorporating real-time data to validate our findings and refine the proposed models further.

Future research should consider specific scenarios, varying consumer distributions, and warehouse locations to maximize the benefits of the hybrid approach. Additionally, future work should focus on proposing and developing different algorithms for planning delivery paths and optimizing warehouse distribution strategies. These efforts will contribute to enhancing the efficiency and sustainability of last-mile parcel delivery systems.

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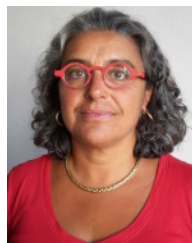
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